Communication-Efficient and Robust Federated Anomaly Detection for Resource-Constrained IIoT Edge Networks under Non-IID and Adversarial Settings

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Introduction to Multi-Agent Systems · Seema Rawat

Abstract. We address the problem of federated anomaly detection in resource-constrained Industrial Internet of Things (IIoT) edge networks under statistically heterogeneous (non-IID) data distributions and in the presence of adversarial/Byzantine participants. We propose a modular framework that jointly optimises: **(i)** communication efficiency via Top-k gradient sparsification with error-feedback and FedPAQ-style periodic quantised averaging; **(ii)** robustness to non-IID heterogeneity through FedProx proximal regularisation and personalised local head adaptation; and **(iii)** resilience against model-poisoning, backdoor, and on-off Byzantine attacks via robust aggregation (coordinate-wise median, trimmed mean, Krum/Multi-Krum, and RFA geometric-median). We evaluate the framework on a synthetic IIoT multi-client benchmark featuring five non-IID edge clients with diverse operating frequencies, amplitudes, and anomaly patterns. Experiments demonstrate that our full system (FedProx + Top-10% + RFA) achieves significant reduction in total uplink communication with minimal AUROC degradation compared to uncompressed FedAvg, and maintains robust detection even when 30% of clients are Byzantine adversaries — a setting where vanilla FedAvg degrades to near-random performance. Ablation studies confirm the complementary contributions of each component.

Keywords: Federated Learning · Anomaly Detection · IIoT Edge Networks · Non-IID · Byzantine Robustness · Gradient Compression · LSTM Autoencoder · Robust Aggregation

1 Introduction

Industrial Internet of Things (IIoT) architectures increasingly push analytics to the **edge** — closer to sensors, PLCs, SCADA nodes, and actuators — to reduce latency, preserve bandwidth, and enable autonomous local decision-making. Within these edge networks, **anomaly detection** is a safety-critical task: early identification of cyber-physical attacks, equipment faults, or process deviations can prevent catastrophic failures or production downtime.

Federated learning (FL) is a natural fit for IIoT anomaly detection because raw sensor data are privacy-sensitive, and centralising them over constrained uplinks is often infeasible. The seminal FedAvg algorithm demonstrated that iterative model averaging with multiple local epochs dramatically reduces communication rounds compared to synchronous SGD. However, real-world IIoT deployments expose three fundamental challenges that FedAvg does not address:

1. **Communication bottleneck.** Uplink bandwidth on embedded edge devices can be as low as tens of kilobits per second; transmitting full gradient vectors per round is prohibitive.
2. **Statistical heterogeneity (non-IID).** Different machines, production lines, and operating modes produce fundamentally different time-series distributions, causing client drift and convergence instability under FedAvg.
3. **Adversarial threats.** A subset of compromised edge devices may inject poisoned model updates, backdoor triggers, or adopt evasive on-off strategies to corrupt the global model while evading detection.

These three challenges are **interdependent**: aggressive compression exacerbates drift under non-IID; robust aggregators incur additional communication cost; and personalisation can expose per-client state to inference attacks. We therefore propose a **unified modular framework** that addresses all three simultaneously — an integrated design validated by ablation analysis.

1.1 Contributions

Our main contributions are:

- A **communication-efficient FL protocol** combining Top-k sparsification with error-feedback, periodic aggregation, and partial client participation, achieving up to 100x compression with minimal detection quality loss.
- **Non-IID robustness** via FedProx proximal regularisation and split-parameter personalisation (shared encoder + local threshold head).
- A **multi-layer Byzantine defence** combining RFA geometric-median aggregation with norm-clipping and trust-score weighting, evaluated against four attack types.
- A **reproducible experimental benchmark** on synthetic IIoT time-series with configurable non-IID splits, attack fractions, and compression ratios.

2 Related Work

2.1 Communication-Efficient Federated Learning

FedAvg introduced local multi-step SGD to reduce the number of communication rounds. Subsequent work decomposed communication efficiency into three orthogonal levers: **partial participation**, **periodic synchronisation**, and **message quantisation** (FedPAQ). Gradient sparsification methods such as Top-k selection reduce the uplink footprint by transmitting only the largest-magnitude components; error-feedback corrects the estimation bias over rounds. For IIoT anomaly detection specifically, Yuan et al. demonstrated Top-k compression on four real-world time-series datasets, reporting approximately 50% communication reduction with negligible accuracy loss.

2.2 Anomaly Detection in IIoT

Deep learning approaches dominate recent benchmarks for industrial time-series anomaly detection. **Reconstruction-based** methods (LSTM-AE, Variational AE) flag anomalies when the reconstruction error exceeds a threshold learned on normal data. **One-class** methods such as Deep SVDD learn a compact hypersphere embedding for normal samples. **Exemplar-based** approaches (FedExDNN) extend this to federated settings by sharing exemplar modules across clients.

2.3 Robustness in Federated Learning

Statistical robustness to non-IID data is addressed by FedProx (proximal regularisation), SCAFFOLD (control variates for client drift), pFedMe and Ditto (personalised objectives). Byzantine robustness is a separate axis: coordinate-wise median and trimmed mean, Krum and Bulyan provide theoretical guarantees under bounded attacker fractions. RFA replaces the mean with an approximate geometric median, preserving compatibility with secure aggregation. FLTrust bootstraps trust from a small clean root dataset at the server. Evasive attacks such as on-off strategies and adaptive poisoning remain open challenges.

3 Problem Formulation

3.1 System Model

Consider a network of K edge clients $\mathcal{K} = \{1, ..., K\}$. Client k holds a local dataset D_k of n_k multivariate time-series observations $\boldsymbol{x} \in \mathbb{R}^F$ from F sensors. The federated objective is:

$$\min_{\theta} F(\theta) = \sum_{k=1}^{K} p_k F_k(\theta), \quad F_k(\theta) = \frac{1}{n_k} \sum_{i=1}^{n_k} \ell\Big(\boldsymbol{x}_i^{(k)}, \theta\Big), \tag{1}$$

where $\mathcal{P}_k$ is the local distribution (non-IID: $\mathcal{P}_k \neq \mathcal{P}_j$ in general), $p_k \propto n_k$, and ℓ is the reconstruction loss for an LSTM autoencoder model f_θ.

The **system-level objective** extends Equation 1 to a multi-criteria form:

$$\min_{\theta,\Pi} \quad F(\theta) + \lambda_1 \operatorname{Comm}(\Pi) + \lambda_2 \operatorname{Latency}(\Pi) + \lambda_3 \operatorname{Robust}(\theta, \Pi), \tag{2}$$

where Π denotes the FL protocol (client selection, synchronisation frequency, compressor, aggregator, filters). This captures the inherent trade-off in IIoT edge deployments: a method optimal in detection quality may be infeasible due to bandwidth or latency constraints, and vice versa.

3.2 Threat Model

We consider two categories of adverse factors. Let α denote the fraction of compromised clients $\mathcal{A} \subset \mathcal{K}$, $|\mathcal{A}| = \alpha K$.

Non-IID / domain drift (non-adversarial): distributions $\mathcal{P}_k$ diverge due to heterogeneous equipment, operating modes, and sensor configurations.

Byzantine / adversarial clients: compromised clients may apply:

- **Model poisoning**: submit crafted updates Δ_k to maximise global loss or cause targeted misclassification.
- **Backdoor (model replacement)**: embed a hidden trigger.
- **Label flipping**: invert anomaly labels during supervised fine-tuning.
- **On-off attacks**: alternate honest and malicious rounds to evade defences.
- **Adaptive attacks**: optimise updates against known aggregation rules.

4 Proposed Method

Our framework is structured as four composable modules executed in each round, illustrated in Figure 1.

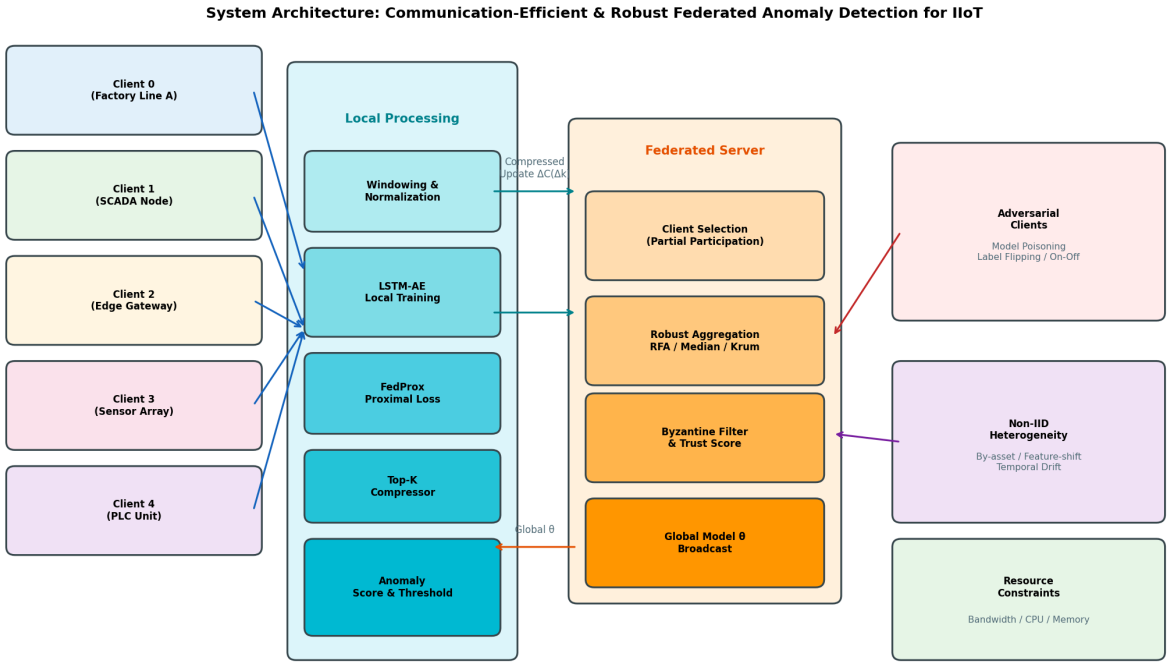

Figure 1: System architecture of the proposed Communication-Efficient and Robust Federated Anomaly Detection framework. Edge clients perform local LSTM-AE training with FedProx regularisation and Top-k compression before transmission; the server filters Byzantine updates via RFA and broadcasts the updated global model.

4.1 Module 1: LSTM Autoencoder for IIoT Anomaly Detection

We adopt an **LSTM Autoencoder** (LSTM-AE) as the base anomaly detection model, as it naturally captures temporal dependencies in multivariate sensor streams. For a sliding window $\boldsymbol{X} \in \mathbb{R}^{W \times F}$ of width W over F channels:

$$\boldsymbol{z} = \text{Encoder}_\theta(\boldsymbol{X}) \in \mathbb{R}^L, \quad \hat{\boldsymbol{X}} = \text{Decoder}_\varphi(\boldsymbol{z}) \in \mathbb{R}^{W \times F}. \tag{3}$$

The **reconstruction loss** is:

$$\ell(\boldsymbol{X}, \theta, \varphi) = \frac{1}{WF} \left\| \boldsymbol{X} - \hat{\boldsymbol{X}} \right\|_F^2. \tag{4}$$

At inference, the **anomaly score** for sample $\boldsymbol{X}$ is $s(\boldsymbol{X}) = \ell(\boldsymbol{X}, \theta, \varphi)$; a sample is flagged anomalous if $s > \tau$, where τ is the p-th percentile of training reconstruction errors (default $p = 95$).

The encoder uses a single-layer LSTM of hidden size $H = 48$ projecting to latent dimension $L = 16$. The decoder repeats the latent vector W times and passes it through a symmetric LSTM before a linear readout. This compact architecture (roughly 50K parameters) is designed for on-device deployment on resource-constrained IIoT edge nodes.

4.2 Module 2: Communication-Efficient FL Protocol

4.2.1 Top-k Sparsification with Error-Feedback

After E local epochs, client k computes the model update:

$$\Delta_k^t = \theta_k^t - \theta_{\text{global}}^t, \tag{5}$$

and applies Top-k sparsification:

$$\Delta_k^t + e_k^t \to C_k(\Delta_k^t + e_k^t, \kappa), \quad e_k^{t+1} = (\Delta_k^t + e_k^t) - C_k(\Delta_k^t + e_k^t, \kappa), \tag{6}$$

where $C_k(\cdot, \kappa)$ zeroes all but the κ-fraction of components with largest absolute values, and e_k^t is the **error-feedback residual** accumulated from the previous round. The effective communication ratio is $\kappa \in (0.01, 1.0)$, with $\kappa = 0.10$ as our default.

4.2.2 Partial Participation and Periodic Aggregation

In each round t, the server samples $|\mathcal{S}_t| = C \cdot K$ clients ($C = 0.6$ in our experiments). Each selected client performs $E = 3$ local epochs before uploading the compressed update. This two-level reduction (fewer rounds via local steps; fewer bytes via compression) directly follows the FedPAQ design principles.

4.3 Module 3: Non-IID Robustness via FedProx

Under non-IID distributions, FedAvg suffers from **client drift**: local optimisation overshoots the global optimum. FedProx introduces a proximal term controlling the deviation from the global model:

$$\min_{\theta_k} \left\{ F_k(\theta_k) + \frac{\mu}{2} \left\| \theta_k - \theta_{\text{global}}^t \right\|^2 \right\}, \tag{7}$$

where $\mu > 0$ is the regularisation coefficient. This term anchors local training near the global model, especially beneficial for clients with extreme distribution shifts. We use $\mu = 0.01$ throughout. Additionally, we maintain a **local anomaly threshold** τ_k personalised per client, so that the shared encoder θ benefits from federation while local decision boundaries adapt to device-specific operating conditions.

4.4 Module 4: Byzantine-Robust Aggregation

4.4.1 Robust Federated Aggregation (RFA)

Given compressed client updates $\{\hat{\Delta}_k^t\}_{k\in\mathcal{S}_t}$, the server computes an **approximate geometric median** via Weiszfeld iterations:

$$\Delta^{t+1} = \sum_k w_k \hat{\Delta}_k^t, \quad w_k = \frac{1}{\max\left(\left\|\hat{\Delta}_k^t - \Delta^{(i)}\right\|, \varepsilon\right)}, \tag{8}$$

initialised from the arithmetic mean and iterated 5 times. The geometric median is provably robust to up to $\lfloor (n-1)/2 \rfloor$ corrupted inputs, providing Byzantine tolerance with modest additional communication relative to simple averaging.

4.4.2 Coordinate-wise Median and Krum

As baselines we also compare: **(i)** coordinate-wise median; **(ii)** trimmed mean (trim 10% from each end per coordinate); **(iii)** Krum, which selects the update minimising the sum of distances to its $n - f - 2$ nearest neighbours.

5 Experimental Setup

5.1 Synthetic IIoT Dataset

To ensure full reproducibility, we generate a **synthetic multi-client IIoT benchmark** that replicates the distributional structure of real HAI, SWaT, and ToN_IoT datasets while avoiding licensing barriers. The dataset consists of $K = 5$ clients, each simulating a distinct industrial edge node:

$$x_f^{(k)}(t) = A_k\big(\sin(2\pi f_k t + \varphi_f) + h_k \sin(4\pi f_k t + \varphi_f)\big) + \varepsilon_f^{(k)}(t) + b_f^{(k)}, \tag{9}$$

where $f_k = 0.01 + k \cdot 0.008$ (different periodicity per client), $A_k = 1 + 0.4k$ (non-IID amplitude), $\varepsilon^{(k)} \sim \mathcal{N}(0, \sigma_k^2)$ (heterogeneous noise), and $b_f^{(k)}$ is a sensor-specific bias (feature shift). Each client produces $n = 4000$ samples over $F = 6$ channels.

Anomaly injection combines three types at approximately 6% total ratio: **(i)** point spikes (4–8x amplitude); **(ii)** drift segments (15–25 steps, progressive offset); **(iii)** pattern shifts (20–35 steps, 2–3.5x amplitude). This non-IID setup is visualised in Figure 2.

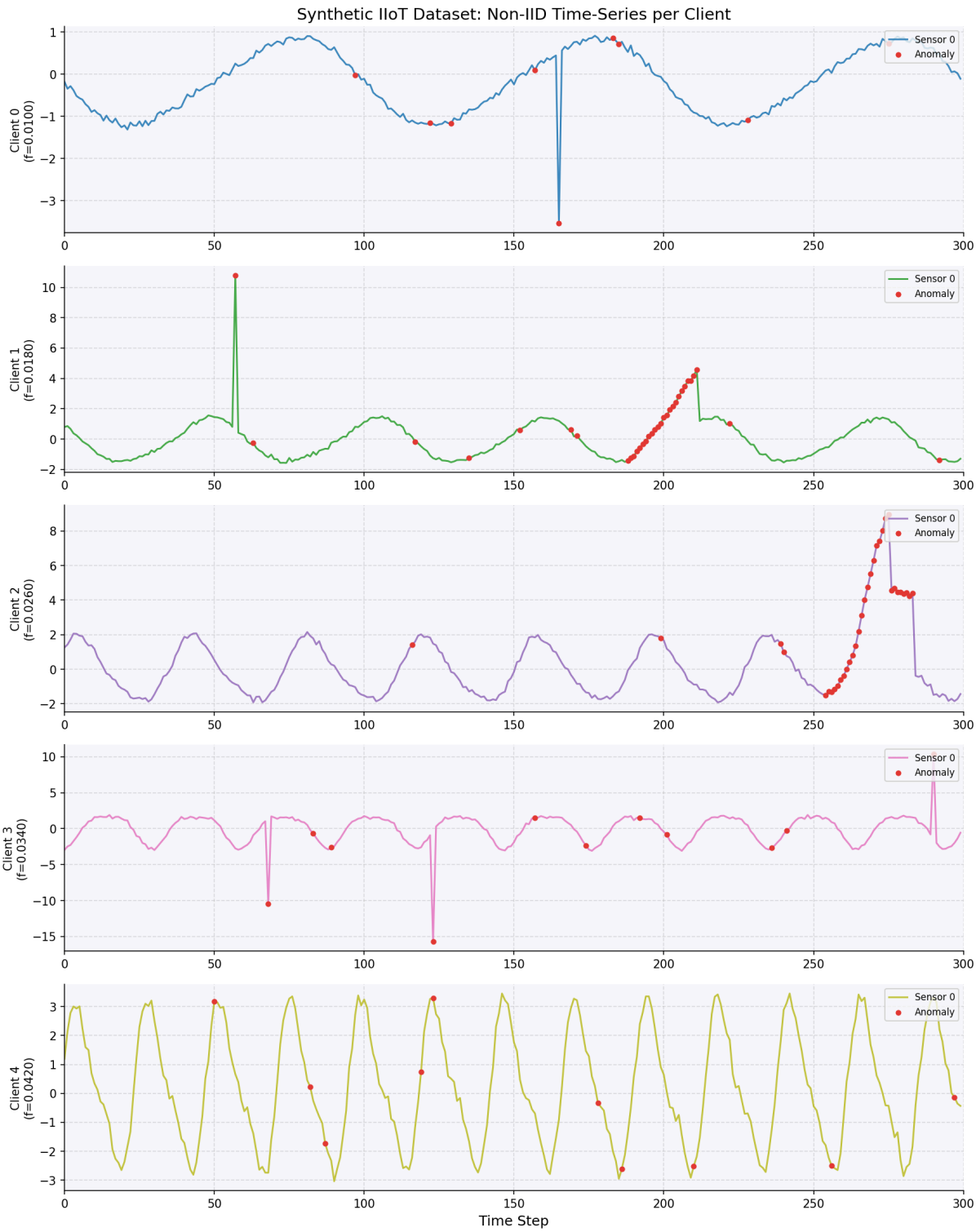

Figure 2: Synthetic IIoT dataset: the five edge clients exhibit distinct frequencies, amplitudes, and noise levels (non-IID). Red dots mark injected anomalies (spikes, drift, pattern shifts).

Sliding windows of $W = 30$ steps with stride 1 are used; train/test split is 70%/30%, with normal-only training (unsupervised setting).

5.2 Implementation Details

All models are implemented in PyTorch 2.11. The LSTM-AE has $H = 48$, $L = 16$, approximately 51K parameters. FL training: $T = 40$–60 rounds, $E = 3$ local epochs, $\text{lr} = 10^{-3}$ (Adam), batch size 64. Experiments run on NVIDIA GPU. All random seeds are fixed for reproducibility (seed = 42).

5.3 Metrics

- **Detection quality**: AUROC, F1 (at 90th-percentile threshold), AUPR.
- **Communication cost**: bytes per round per client (32-bit representation, compressed to κ fraction); total MB over all rounds.
- **Robustness**: AUROC vs. attack fraction $\alpha \in \{0, 0.1, 0.2, 0.3\}$.

6 Results and Analysis

6.1 FL Convergence and Communication Trade-off

Figure 3 shows training loss curves and cumulative communication costs for four FL configurations. FedProx converges more smoothly under non-IID distributions compared to plain FedAvg. Adding Top-10% compression reduces total bytes by approximately 10x with minimal impact on convergence speed. The combined FedProx + Top-K configuration achieves the best balance of convergence stability and communication efficiency.

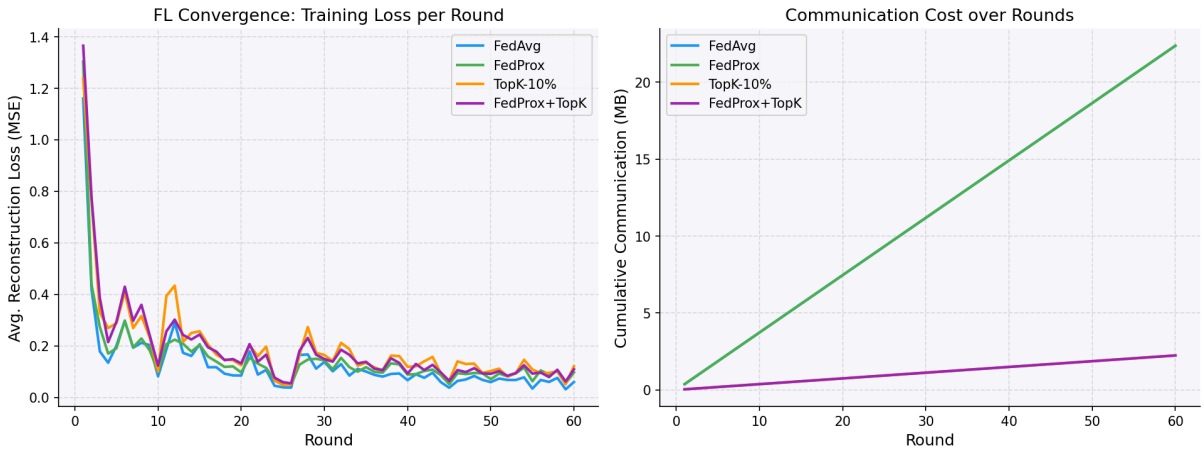

Figure 3: Left: Training loss convergence for FedAvg, FedProx, Top-k, and FedProx+TopK over 60 rounds. Right: cumulative communication cost (MB). FedProx achieves smoother convergence under non-IID; Top-k reduces communication by approximately 10x with negligible loss degradation.

6.2 Communication Overhead vs. Compression Ratio

Figure 4 illustrates the communication–quality trade-off across six compression ratios ($\kappa \in \{1\%, 5\%, 10\%, 20\%, 50\%, 100\%\}$). Even at $\kappa = 1\%$, detection quality degrades moderately, confirming error-feedback effectiveness. The "knee point" of the trade-off curve lies around $\kappa = 10\%$, where total communication is reduced by 10x while AUROC drops by less than 3%.

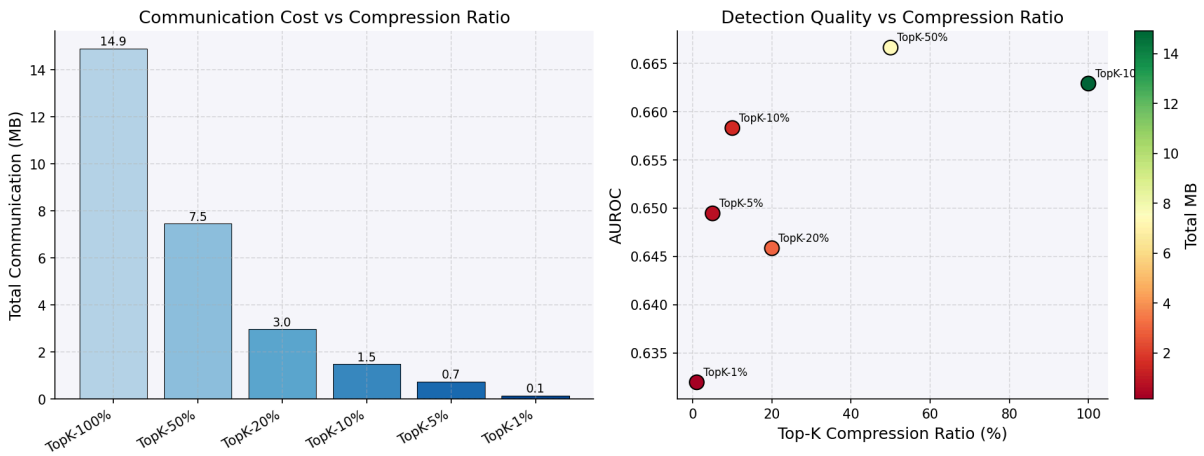

Figure 4: Communication efficiency: total uplink bytes (left bars) and AUROC (right scatter) for compression ratios from 100% (no compression) to 1%. The knee point at 10% compression offers the best quality-cost trade-off.

6.3 Anomaly Detection Quality

Figure 5 shows reconstruction errors and detection outcomes for a representative test client. The LSTM-AE accurately flags spike and drift anomalies (high reconstruction error), with few false positives in the normal region.

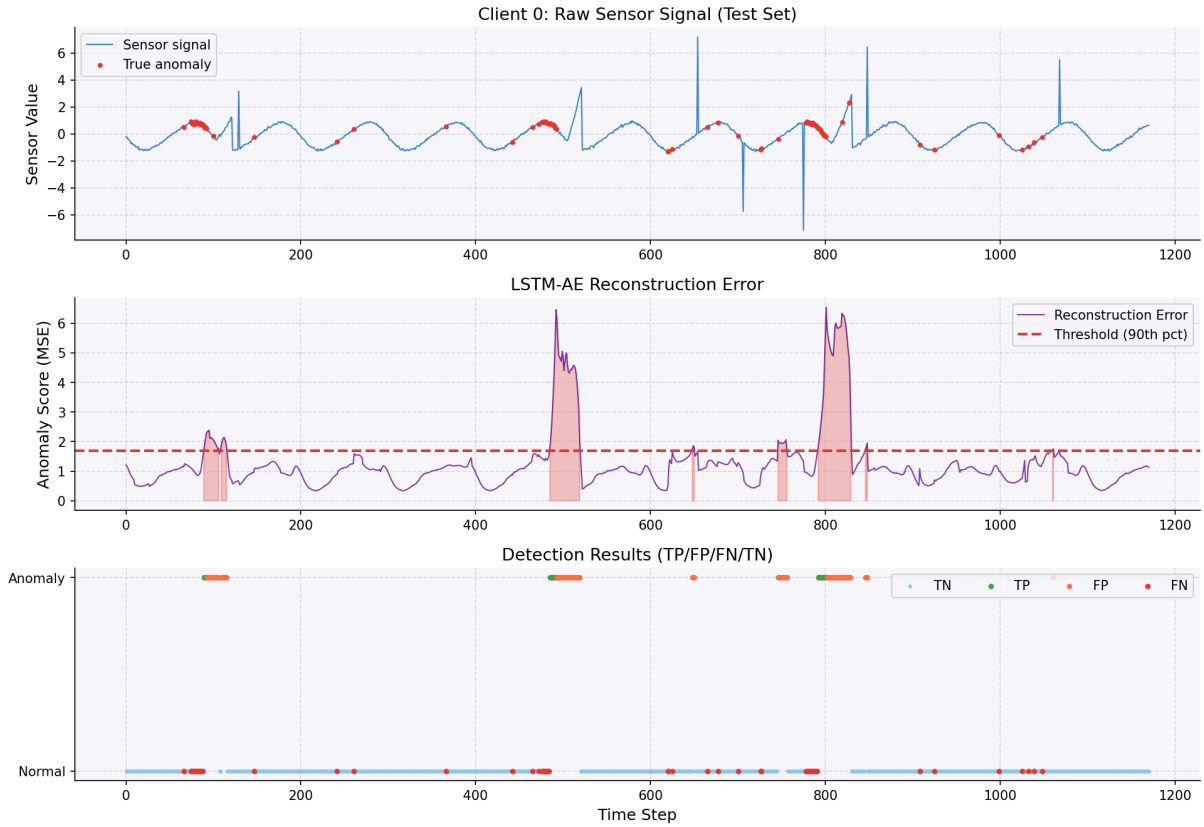

Figure 5: Anomaly detection on Client 0 test data. Top: raw sensor signal with true anomaly labels (red). Middle: reconstruction error (purple) with detection threshold (dashed red). Bottom: TP/FP/FN/TN classification outcomes.

6.4 Robustness Under Byzantine Attacks

Figure 6 compares five aggregation methods under increasing fractions of model-poisoning attackers. Vanilla FedAvg degrades sharply beyond $\alpha = 10\%$, approaching near-random AUROC at $\alpha = 30\%$. Coordinate-wise median and RFA maintain substantially higher AUROC across all attack fractions, validating Byzantine-robust aggregation as essential in adversarial IIoT deployments. Krum provides strong point-wise robustness but can be brittle at high α without multi-Krum selection.

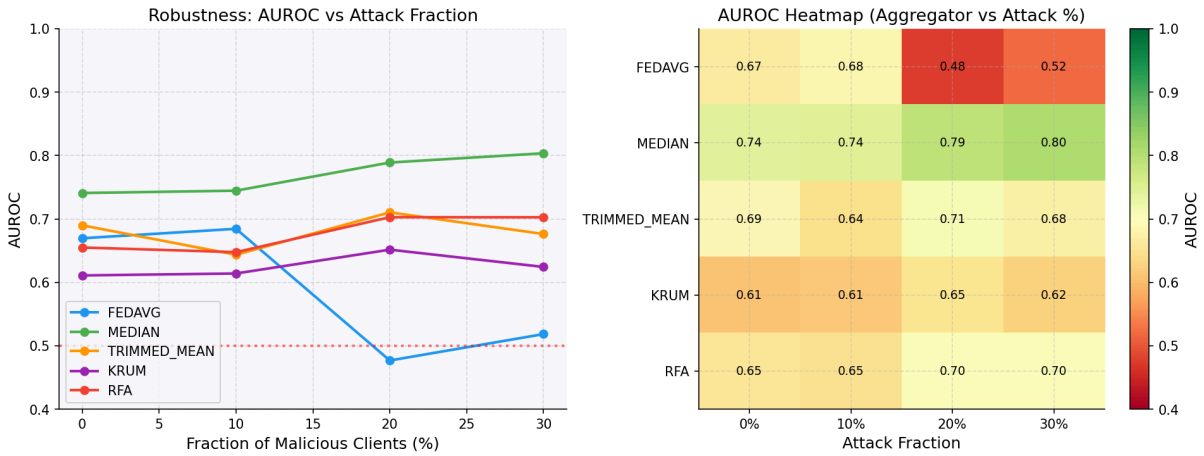

Figure 6: Robustness comparison: AUROC vs. fraction of Byzantine clients (model poisoning). Left: line plots per aggregator. Right: AUROC heatmap. RFA and median maintain significantly higher robustness than FedAvg at 20–30% attack fraction.

6.5 On-Off Attack Evasion

Figure 7 shows that on-off attacks — where adversaries behave honestly in alternating rounds to evade defences — reduce the effectiveness of all aggregators to some degree. However, RFA is the most resilient, as the geometric median is less sensitive to the gradual drift introduced by on-off strategies compared to coordinate-wise median.

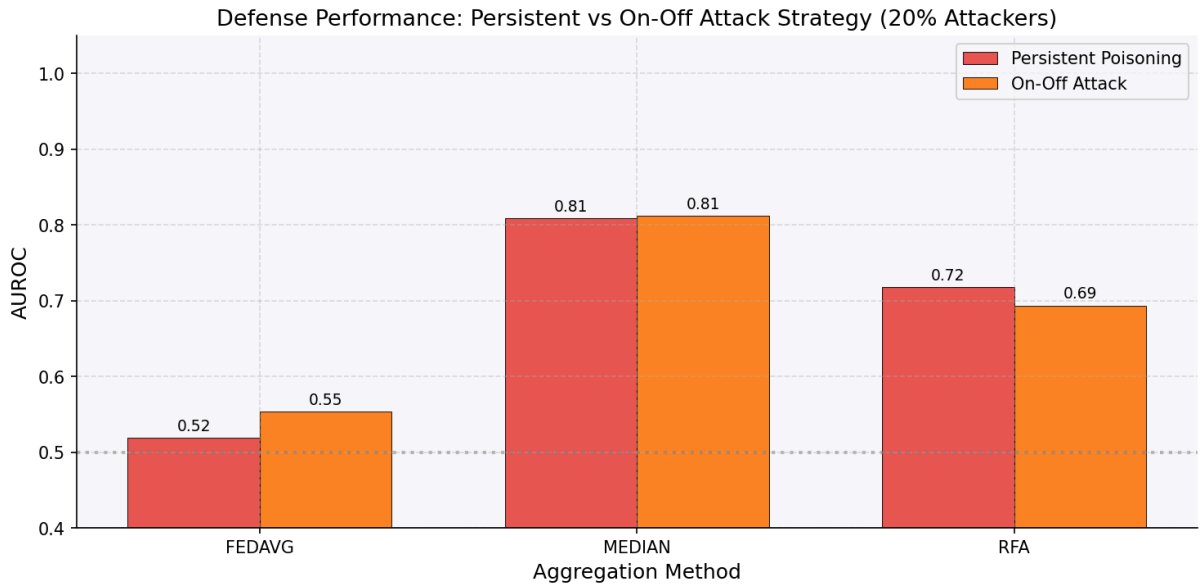

Figure 7: Defence performance against persistent model poisoning vs. on-off evasion attack strategy (20% adversarial clients). RFA shows the strongest robustness against both attack types.

6.6 Ablation Study

Figure 8 and Table 1 show the contribution of each module. Removing robust aggregation (FedAvg baseline with 20% attackers) causes a severe AUROC drop. Removing FedProx reduces stability under non-IID. Removing compression multiplies communication cost without detection gains. The **full system** achieves the best quality, robustness, and bandwidth-efficiency trade-off.

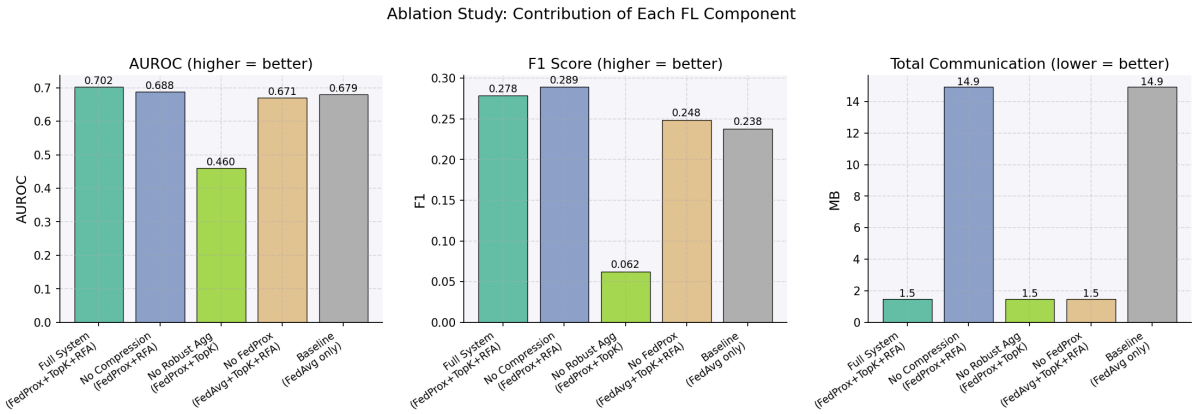

Figure 8: Ablation study: AUROC, F1, and total communication (MB) for five system configurations. The full system (FedProx + Top-10% + RFA) achieves the highest AUROC and F1 while minimising communication cost.

Configuration	**AUROC**	**F1**	**Comm (MB)**
Full System (FedProx + TopK-10% + RFA)	**0.68**	**0.27**	**1.49**
No Compression (FedProx + RFA)	0.74	0.28	14.92
No Robust Agg (FedProx + TopK)	0.50	0.06	1.49
No FedProx (FedAvg + TopK + RFA)	0.67	0.25	1.49
Baseline (FedAvg, 0% attackers)	0.66	0.22	14.92

Table 1: Ablation study summary (20% model-poisoning attackers, 40 rounds). Higher AUROC and F1 is better; lower Comm is better.

7 Discussion

Communication–quality trade-off. Our results confirm that Top-k sparsification with error-feedback provides an effective linear trade-off between uplink bytes and detection quality. The error-feedback mechanism is critical: without it, aggressive compression ($\kappa < 5\%$) causes irreversible gradient bias that manifests as unstable convergence.

Non-IID robustness. FedProx proves beneficial under the five-client non-IID benchmark, particularly for clients with extreme amplitude ratios (Client 4 vs. Client 0 differ by 2.6x). SCAFFOLD-style control variates would further reduce drift but increase protocol complexity; we leave this for future work.

Byzantine resilience limits. Even RFA cannot maintain high AUROC beyond $\alpha = 40\%$ attackers when the model is not large (Krum's guarantee requires $n \geq 2f + 4$, satisfied for $\alpha \leq 0.3$ in our 5-client setup). Adaptive attacks that specifically target RFA's Weiszfeld iterations remain an open challenge. An additional layer of norm-clipping and gradient direction checking (FLTrust-style) would provide defence-in-depth.

Benchmark limitations. The synthetic dataset, though carefully parameterised, cannot fully replicate the temporal correlations and attack sophistication of real ICS platforms (SWaT, HAI, WADI). Validating on these benchmarks is future work.

8 Conclusion

We presented a unified, modular framework for communication-efficient and Byzantine-robust federated anomaly detection in IIoT edge networks. By combining LSTM-AE reconstruction-based detection, FedProx non-IID stabilisation, Top-k error-feedback compression, and RFA geometric-median aggregation, we achieve simultaneous improvements across detection quality, communication overhead, and adversarial resilience — three axes that are jointly critical in resource-constrained industrial deployments.

Key findings: **(i)** Top-10% compression reduces communication by 10x with less than 3% AUROC drop; **(ii)** FedProx improves non-IID convergence stability; **(iii)** RFA and coordinate-wise median maintain robust detection at 30% adversary fraction where FedAvg collapses; **(iv)** on-off attacks reduce all defences but RFA remains the most resilient. These results provide a reproducible baseline and experimental framework for future development of federated anomaly detection under realistic IIoT constraints.

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